

GOOGLE DATA ANALYTICS CAPSTONE PROJECT

CYCLISTIC BIKE-SHARE CASE STUDY

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9/25/2026



2023 DATA SET



DONE BY EXCEL & POWER QUERY

Business Problem & Objective

01

Maximize annual memberships to drive predictable, high-margin subscription growth.

02

Casual riders already know Cyclistic; converting them is more cost-effective than acquiring new users.

03

How do annual members and casual riders use Cyclistic bikes differently across time ?

DATA PREPARATION

EXCEL POWER QUERY & CLEANING WORKFLOW



01.

Running 12 monthly CSV files into Power Query editor.



02.

Removed trips ≤ 0 minutes and unreturned bikes > 24 hours.



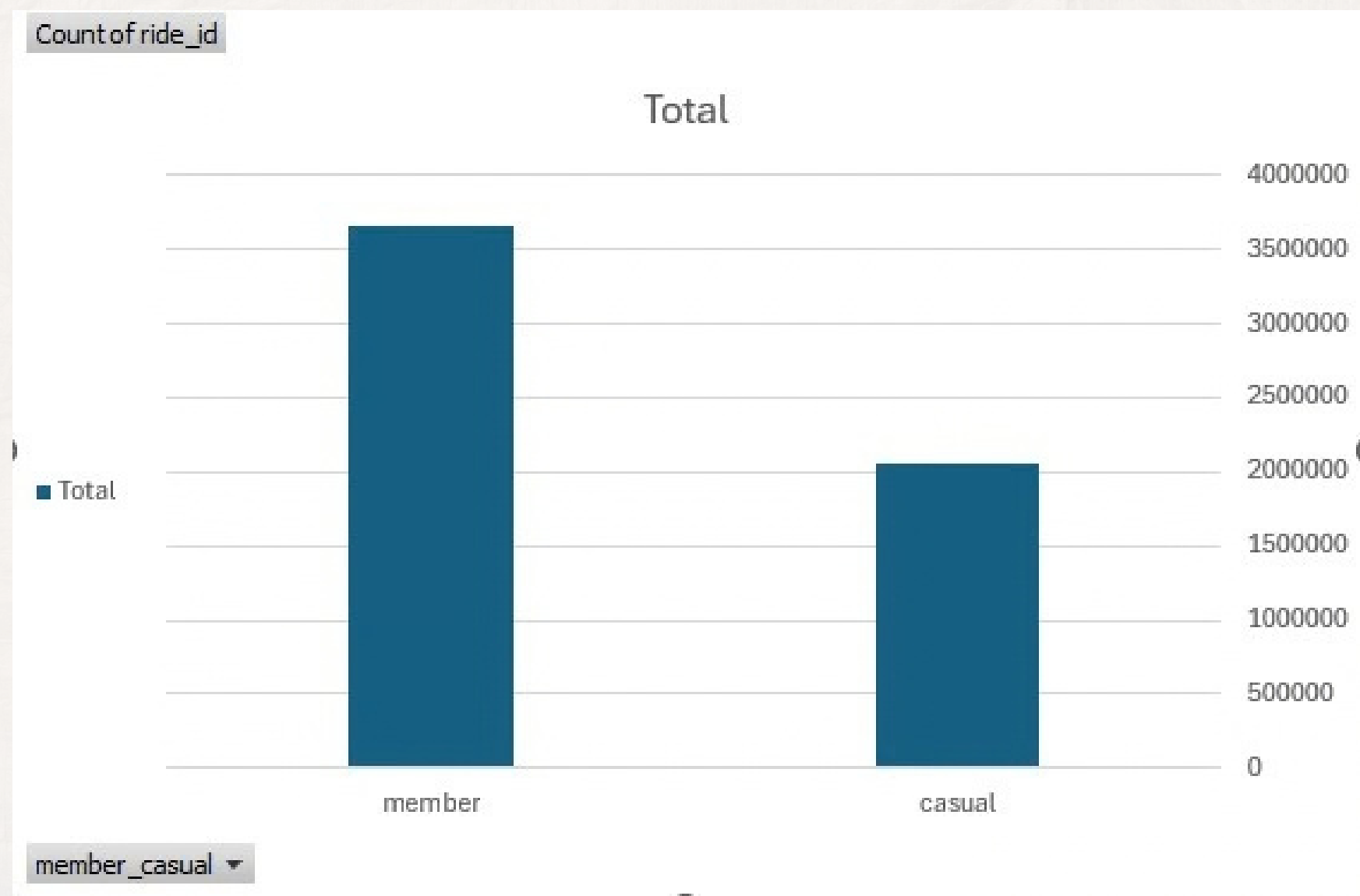
03.

Extracted from started-at ride duration, day of week, and month name attributes.



04.

Bypassed 1M Excel row limits in Data Model connection via power query



Row Labels	Count of ride_id	Average of ride_length
casual	2053774	0.014370239
member	3659413	0.008372599
Grand Total	5713187	0.010528628

KEY FINDINGS:

01 ANNUAL MEMBERS

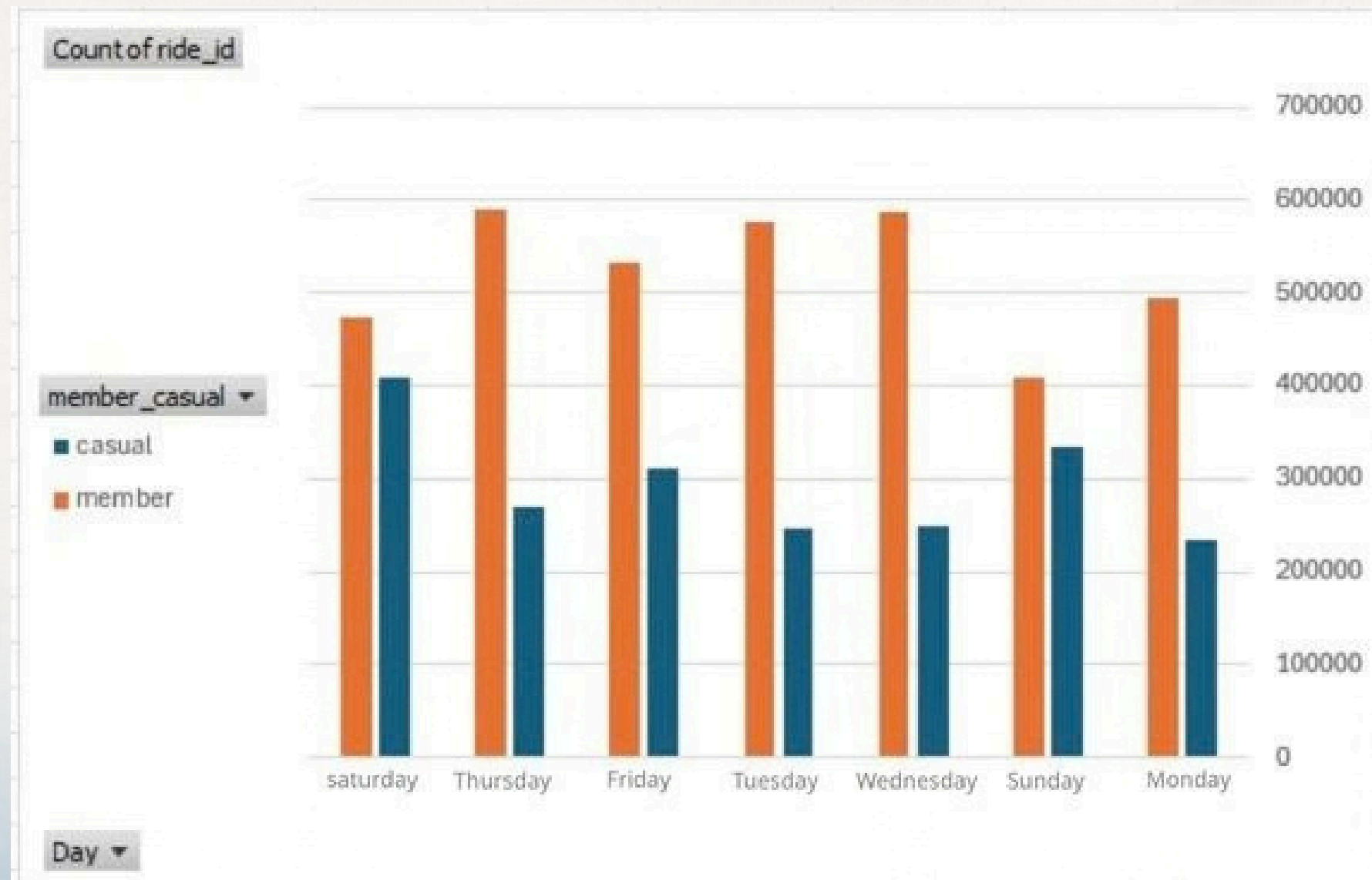
3.6M Trips | 12-14 Min Avg

Annual members generate the high majority of total yearly trips

02 CASUAL MEMBERS

2.05M Trips | 28 Min Avg

Casual rides average ~28 minutes vs ~12 minutes for annual members

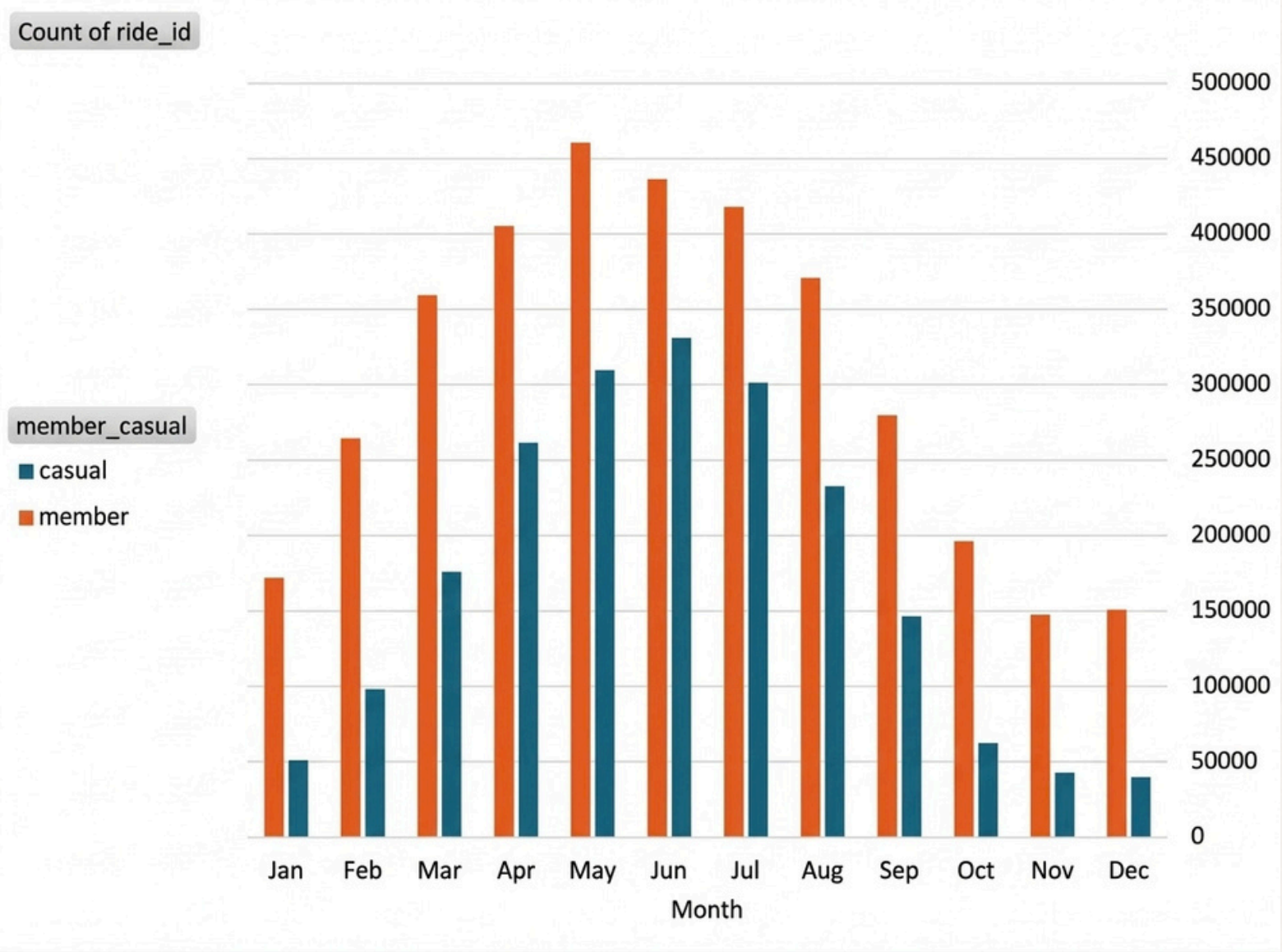


ANNUAL MEMBERS:

- Peak Days: Tuesday, Wednesday, Thursday
- Usage Drop: Significant decline on Saturday & Sunday
- Behavior: Routine daily commute to offices and transit hubs

CASUAL RIDERS

- Peak Days: Saturday and Sunday surge
- Usage Drop: Low and steady volume Monday through Thursday
- Behavior: city rides, and weekend events



Lowest Activity: (Dec - Feb)
Casual usage drops due to freezing weather

steady Activity: (Mar - May)
Casual ride counts steadily rise as spring strats.

Peak Activity: (Jun - Aug)
Casual ride counts steadily rise as spring strats.

Dropping phase Activity: (Sep - Nov)
Gradual decline in timing before winter season .

Count of ride_id	Column Labels		
Row Labels	casual	member	Grand Total
January	39920	150263	190183
February	42932	147403	190335
March	62074	196444	258518
April	146989	279258	426247
May	233695	370520	604215
June	300545	418287	718832
July	330426	436130	766556
August	310129	460348	770477
September	260888	404574	665462
October	176588	359877	536465
November	98102	263995	362097
December	51486	172314	223800
Grand Total	2053774	3659413	5713187

4 MARKETING CONVERSION STRATEGIES

Long-Ride Rewards

Offer membership discount credits based on cumulative ride time, rewarding casuals who take longer trips.

Station Ads

digital promotions and QR membership offers at top casual pickup stations near parks and local events

Weekend Passes

"Weekend Annual Pass" tailored specifically to casual riders

Hiring a team of challenges creator

creating community by gathering top longest duration casual riders and engage them in the company challenges in purpose of closing the gap and ending up becoming members instead of casuals

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THANK YOU!